

Towards Cross Reality: A Comparison of World-in-Miniature and Virtual Reality Switch

Damla Welte^{*}
ETH Zurich

Mathieu Lutfallah[‡]
ETH Zurich

Long Cheng[§]
ETH Zurich & RhySearch

Andreas Kunz[¶]
ETH Zurich

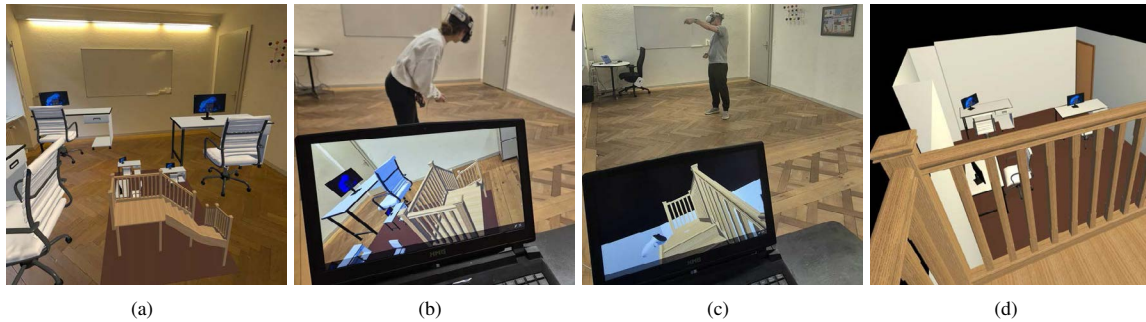


Figure 1: Screenshots of users testing the environment: (a) Mixed Reality is enhanced with a World-in-Miniature model. (b) The user brings her head close to the staircase within the World-in-Miniature model to observe the room from a scaled-down perspective in Mixed Reality. (c) After switching to Virtual Reality, the user employs teleportation to navigate up the staircase. (d) The user observes the room from the top of the staircase at an accurate perspective in Virtual Reality.

ABSTRACT

Visualization of complex architectural and interior design proposals demands spatial comprehension. Mixed Reality (MR) has opened new ways for architectural visualization through its ability to overlay virtual models onto already existing real environments. However, this approach alone restricts users to the physical constraints of their surroundings. We developed two approaches to augment MR with additional perspectives from physically inaccessible viewpoints: the World-in-Miniature (WIM) solution and the Cross Reality (CR) solution. The WIM solution provides a scaled-down 3D replica of the virtual environment, whereas the CR solution enables users to switch to Virtual Reality and teleport to physically inaccessible viewpoints. Our research aims to explore whether one solution outperforms the other.

Index Terms: Cross Reality, Mixed Reality, Virtual Reality, Spatial Comprehension, World-in-Miniature

1 INTRODUCTION

Spatial comprehension is a skill required across various disciplines, from architecture and urban planning to medicine, art, and engineering, enabling professionals to visualize and interact with 3D spaces. This skill typically develops through years of experience. Tools like 3D models and digital visualizations aid in bridging gaps for those with limited spatial comprehension, such as clients of architects. Mixed Reality (MR) devices, such as optical and video see-through head-mounted displays (HMDs), enhance architectural visualization by overlaying digital designs onto physical

spaces, while navigation in MR is constrained by physical movement within the user's environment. To address 3D perspective challenges and enhance spatial exploration, we designed two solutions through a spatial perspective-taking task: (i) the World-in-Miniature (WIM) solution, which augments MR with a virtual miniature model of the environment, and (ii) the Cross Reality (CR) solution, which enables transitions between MR and Virtual Reality (VR). The use of WIM is well-explored in the literature and established as an effective method for visualizing and interacting with spaces. We explore whether CR can achieve the same benefits and potentially offer additional advantages, such as providing a full-scale view from other perspectives while mitigating the overhead associated with using a more complex system.

A WIM allows examining scaled-down designs from any angle, offering comprehensive spatial insights, though it requires cognitive effort to translate dimensions into real-world scale. Bruder & Steinicke [1] integrated transitions from the WIM to full-scale VR through portals. Yet, their system was confined to VR settings. Ihara & Kawaguchi [4] showed the feasibility of WIM in MR. However, they focused on object placement and alignment rather than understanding the benefits of spatial viewing. CR, particularly transitioning from MR to VR, offers navigation options like flying or teleportation to explore physically inaccessible areas. Eissele et al. [3] showed that CR reduced task completion times and error rates compared to 2D drawings. However, these findings were based on a small sample size, indicating the need for more studies. Roo et al. [5] proposed that using MR and VR with teleportation could enhance spatial perception. Cho et al. [2] studied an object finding and placing task, where CR outperformed MR alone in usability, completion speed, and comprehension of the environment. In CR, specifically while in VR, users could scale down the room virtually and achieve a bird's-eye view to better understand the scene, a feature akin to a WIM in MR. The specific reasons for CR's advantages over other techniques, whether due to the bird's-eye view or the integration of real-scale perspectives in VR, remain unclear. If a bird's-eye view is sufficient, switching to VR becomes unnecessary, given that a WIM model suffices for obtaining such a view and is feasible within MR, as shown by Ihara & Kawaguchi [4].

^{*}Contributed Equally

[†]e-mail: damla.welte@gmail.com

[‡]e-mail: lutfallah@iwf.mavt.ethz.ch

[§]e-mail: long.cheng@iwf.mavt.ethz.ch

[¶]e-mail: kunz@ethz.ch

2 IMPLEMENTATION

We developed a standalone application on the Meta Quest 3 to augment spatial comprehension in a spatial planning task. The task involves strategically placing privacy partitions in an office to obscure as many viewpoints of computer monitors as possible from a staircase. To emulate the office setup, the virtual environment (VE) features a pre-arranged, fixed layout consisting of a staircase and three workstations, each equipped with a desk, chair, and computer monitor. We created two distinct scenes to mitigate learning effects in a future within-person study comparing the two solutions. The main differences between these scenes are the shape of the staircase and the arrangement of the workstations. In addition, a simpler one with two desks and a small staircase was created to familiarize users with the system, test its functionalities, and become acquainted with the controller buttons.

To align the VE with the physical room, we use Meta's *Scene Model*. The *Scene Model* is generated by scanning the physical room with the Meta Quest 3 HMD and consists of a collection of *Scene Anchors*. Upon launch, the application retrieves the *Scene Anchor* labeled "FLOOR", adjusts the VE's rotation to match the orientation of the FLOOR anchor and places the VE at the anchor's position. Our application continuously checks and realigns the VE as needed. The FLOOR's dimensions, determined by querying its bounding box, were initially used to set the VE size in Unity. This process ensures that the VE is always centered within any scanned physical room, regardless of its size, and oriented to align with the room's longer and shorter sides.

For general applicability, switching from MR to VR generates a virtual replica of the physical room in real-time, regardless of where the application is launched. By utilizing the HMD to scan the physical room and overlaying it with virtual materials, the virtual floor, walls, and door are aligned precisely with their physical counterparts. We intentionally did not replicate the ceiling in VR, preventing the staircase from being visually truncated and allowing users to ascend the staircase by teleportation beyond the ceiling to view the room from above. The application can also be easily modified to include additional elements, such as the ceiling, tables, couches, windows, and storage, which would then be visualized in VR. This is especially useful for complex, large-scale architectures, as it reduces the effort to model a digital twin of the space.

Our application logic runs as follows: Upon launching the application, users enter the MR setting, where they see the virtual staircase and workstations overlaid onto the physical room at an accurate full-scale. They navigate the environment through real walking. Apart from visual inspection via the HMD, users use controllers to spawn and relocate the privacy partitions. Partitions can only be spawned on the floor, without collision with workstations or the staircase. Once a partition is spawned in an allowed location, users can adjust position and orientation using the controller's grab button. A maximum of five partitions can be spawned.

World-in-Miniature Solution: In the WIM solution, users remain in the MR setting throughout the task. The main feature of this solution is the WIM model: a 1:10 scaled-down virtual twin of the VE with the floor made visible using a brown-colored material (Fig. 1(a)). The WIM is positioned at the center of the room, inspired by Bruder & Steinicke [1], and 0.9 meters above the ground. The scale and position of the model are fixed. The WIM consistently mirrors the full-scale VE. Whenever a partition is spawned or relocated in the full-scale model, its position and rotation in the WIM are updated accordingly through simple coordinate transformations. This allows users to visualize what they would see from the staircase (Fig. 1(b)). Placing partitions is not possible in the WIM model but only in the full-scale VE. This design choice is made to emphasize the augmentation of MR with additional perspectives using the WIM for visualization, rather than to simplify the task by reducing the time or the need for walking.

Cross Reality Solution: The key feature of the CR solution is the ability to switch between realities, from MR to VR and back. In MR setting, users can see the physical room with the VE overlaid at full-scale. Switching to the VR setting is achieved by pressing the "X" button on the left controller, which automatically renders virtual walls (including the door) and the floor. In the VR setting, users can navigate the room either by walking or by using the left joystick to point to specific locations and teleport there (Fig. 1(c)). The teleportation functionality allows access to areas that are unreachable by natural walking, such as the top of the virtual staircase that does not exist in the physical room. From these teleported locations, users can view the space at full-scale (Fig. 1(d)). To ensure a fair comparison with the WIM solution in future evaluations, users are restricted from spawning or moving partitions in the VR setting; they can only use VR to visualize the space. To switch back to the MR setting, where they can spawn and move partitions, users press the "X" button on the controller again.

Measurement: The effectiveness of partition placements is assessed by casting rays from the computer monitors toward the staircase at the end of the task. A task success score is calculated as the ratio of blocked rays to the total number of rays casted. The score can range from zero, indicating that the participant did not cover any rays, to 100, indicating complete coverage of all rays. In addition, the number of partition relocations, switches between MR and VR, and the number of teleportations are tracked. Task completion time is recorded from when the user presses "Start" button on a menu interface and to when the user pressed the "End" button. After pressing the "End" button, the user observes the placement result with ray intersection.

Initial test runs showed both solutions were feasible and well-received, with positive feedback from users. A larger-scale study is planned to evaluate the CR and WIM solutions against each other based on the metrics defined above. Subjective questionnaires will be included to assess system usability and user preference.

3 CONCLUSION

We present two solutions in one standalone software application to augment MR by providing additional perspectives for viewing the VE from inaccessible viewpoints. This enables a comparison of the two prototypes through a human factors study.

REFERENCES

- [1] G. Bruder, F. Steinicke, and K. H. Hinrichs. Arch-explore: A natural user interface for immersive architectural walkthroughs. In *3DUI - IEEE Symposium on 3D User Interfaces 2009 - Proceedings*, pp. 75–82, 2009. doi: 10.1109/3DUI.2009.4811208 1, 2
- [2] H. Cho, B. Yuan, J. D. Hart, Z. Chang, J. Cao, E. Chang, and M. Billingham. Time Travellers: An Asynchronous Cross Reality Collaborative System. In *IEEE International Symposium on Mixed and Augmented Reality Adjunct, ISMAR-Adjunct*, pp. 848–853. Institute of Electrical and Electronics Engineers Inc., 2023. doi: 10.1109/ISMAR-Adjunct60411.2023.00186 1
- [3] M. Eissele, O. Siemoneit, and T. Ertl. Transition of Mixed, Virtual, and Augmented Reality in Smart Production Environments - An Interdisciplinary View. In *Proceedings of the Conference on Robotics, Automation and Mechatronics*, pp. 1–6. IEEE, 2006. doi: 10.1109/RAMECH.2006.252671 1
- [4] K. Ihara and I. Kawaguchi. Virtual Object Placement in MR Space Using a 3D Miniature Model of a Room. In *Conference on Human Factors in Computing Systems*, pp. 23–29. Association for Computing Machinery, 4 2022. doi: 10.1145/3516492.3558789 1
- [5] J. S. Roo, J. Basset, P.-A. Cinquin, and M. Hachet. Understanding Users' Capability to Transfer Information between Mixed and Virtual Reality: Position Estimation across Modalities and Perspectives. In *Proceedings of the Conference on Human Factors in Computing Systems, CHI*, pp. 1–12, 2018. doi: 10.1145/3173574.3173937 1